

EX NO:4 A

SUB QUERIES

DATE:

AIM:

To implement a subqueries for department and employee.

PROCEDURE:

1) Create table with the following attributes

CREATE DATABASE CompanyDB;

USE CompanyDB;

Table: departments

```
CREATE TABLE departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(50),  
    location_id INT,  
    city VARCHAR(50)
```

```
);
```

Table: employees

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    salary DECIMAL(10,2),  
    department_id INT,  
    FOREIGN KEY (department_id) REFERENCES departments(department_id)  
);
```

```
INSERT INTO departments VALUES
```

```
(10, 'HR', 17000, 'New York'),
```

```
(20, 'IT', 18000, 'Chicago'),
```

```
(30, 'Finance', 17000, 'New York'),
```

```
(40, 'Marketing', 19000, 'Boston');
```

```
select*from departments;
```

	department_id	department_name	location_id	city
▶	10	HR	17000	New York
	20	IT	18000	Chicago
	30	Finance	17000	New York
	40	Marketing	19000	Boston
*	NULL	NULL	NULL	NULL

INSERT INTO employees VALUES

(1, 'John', 'Smith', 50000, 10),

(2, 'Alice', 'Brown', 60000, 20),

(3, 'Robert', 'King', 55000, 10),

(4, 'Mary', 'Scott', 70000, 30),

(5, 'David', 'Lee', 45000, 20),

(6, 'Sophia', 'Turner', 80000, 30),

(7, 'James', 'Hall', 40000, 40);

SELECT * FROM employees;

	employee_id	first_name	last_name	salary	department_id
▶	1	John	Smith	50000.00	10
	2	Alice	Brown	60000.00	20
	3	Robert	King	55000.00	10
	4	Mary	Scott	70000.00	30
	5	David	Lee	45000.00	20
	6	Sophia	Turner	80000.00	30
	7	James	Hall	40000.00	40
*	NULL	NULL	NULL	NULL	NULL

2) Find Employees Earning More Than the Average Salary

SELECT first_name, last_name, salary

FROM employees

WHERE salary > (SELECT AVG(salary) FROM employees);

	first_name	last_name	salary
▶	Alice	Brown	60000.00
	Mary	Scott	70000.00
	Sophia	Turner	80000.00

3) Find Employees with Departments with location id 17000

SELECT first_name, last_name

FROM employees

WHERE department_id IN (

SELECT department_id

FROM departments

WHERE location_id = 17000

);

	first_name	last_name
▶	John	Smith
	Robert	King
	Mary	Scott
	Sophia	Turner

4) Find Employees with Salaries Above Their Department's Average

SELECT first_name, salary

FROM employees e1

WHERE salary > (

SELECT AVG(salary)

FROM employees e2

WHERE e1.department_id = e2.department_id

);

	first_name	salary
▶	Alice	60000.00
	Robert	55000.00
	Sophia	80000.00

5) Find Departments with Employees Earning More Than the Average Salary

SELECT department_id, department_name

FROM departments

WHERE department_id IN (

SELECT department_id

FROM employees

WHERE salary > (SELECT AVG(salary) FROM employees)

);

	department_id	department_name
▶	20	IT
	30	Finance
*	NULL	NULL

6) Retrieve Employees and Their Salary Difference from the Average Salary

SELECT first_name, last_name,

salary - (SELECT AVG(salary) FROM employees)

AS salary_difference

FROM employees;

	first_name	last_name	salary_difference
▶	John	Smith	-7142.857143
	Alice	Brown	2857.142857
	Robert	King	-2142.857143
	Mary	Scott	12857.142857
	David	Lee	-12142.857143
	Sophia	Turner	22857.142857
	James	Hall	-17142.857143

7) Find employees whose salaries are above the average salary in their respective departments.

SELECT first_name, last_name, salary

FROM employees e1

WHERE salary > (

SELECT AVG(salary)

FROM employees e2

WHERE e1.department_id = e2.department_id

);

	first_name	last_name	salary
▶	Alice	Brown	60000.00
	Robert	King	55000.00
	Sophia	Turner	80000.00

8) Retrieve departments with an average salary greater than the overall average salary.

SELECT department_id, AVG(salary) AS average_salary

FROM employees

GROUP BY department_id

HAVING AVG(salary) > (SELECT AVG(salary) FROM employees);

	department_id	average_salary
▶	30	75000.000000

9) List employees who belong to departments located in New York

SELECT first_name, last_name

FROM employees

WHERE department_id IN (

```
SELECT department_id  
FROM departments  
WHERE city = 'New York'  
);
```

	first_name	last_name
▶	John	Smith
	Robert	King
	Mary	Scott
	Sophia	Turner

RUBRICS:

OBSERVATION	
RECORD	
TOTAL	

RESULT:

Thus a program to implement a sub queries for employee and department is written and executed successfully

EX NO:4 B

SUB QUERIES

DATE:

AIM:

To implement a subqueries for department and employee.

PROCEDURE:

CREATE DATABASE CompanyDB;

USE CompanyDB;

1) Create DEPARTMENT table

CREATE TABLE DEPARTMENT (

DCODE INT PRIMARY KEY,

DNAME VARCHAR(30)

);

2. Create EMPLOYEE table

CREATE TABLE EMPLOYEE (

ENO INT PRIMARY KEY,

NAME VARCHAR(20),

DOB DATE,

GENDER CHAR(1),

DCODE INT,

FOREIGN KEY (DCODE) REFERENCES DEPARTMENT(DCODE)

);

3. Insert values into EMPLOYEE and DEPARTMENT table.

INSERT INTO DEPARTMENT VALUES (201, 'COMPUTER SC');

INSERT INTO DEPARTMENT VALUES (202, 'INFN SC');

INSERT INTO DEPARTMENT VALUES (203, 'CSE');

INSERT INTO DEPARTMENT VALUES (204, 'MECHANICAL');

select*from DEPARTMENT;

	DCODE	DNAME
▶	201	COMPUTER SC
	202	INFN SC
	203	CSE
	204	MECHANICAL
*	NULL	NULL

INSERT INTO EMPLOYEE VALUES (12345, 'HAMEN', '2001-03-24', 'M', 201);

INSERT INTO EMPLOYEE VALUES (12346, 'VINI', '2001-03-12', 'F', 202);

INSERT INTO EMPLOYEE VALUES (12347, 'ANI', '1999-01-11', 'F', 203);

INSERT INTO EMPLOYEE VALUES (12348, 'PETER', '2001-02-14', 'M', 201);

select*from EMPLOYEE;

	ENO	NAME	DOB	GENDER	DCODE
▶	12345	HAMEN	2001-03-24	M	201
	12346	VINI	2001-03-12	F	202
	12347	ANI	1999-01-11	F	203
	12348	PETER	2001-02-14	M	201
*	NULL	NULL	NULL	NULL	NULL

4. CARTESIAN PRODUCT

SELECT *

FROM EMPLOYEE, DEPARTMENT;

	ENO	NAME	DOB	GENDER	DCODE	DCODE	DNAME
▶	12348	PETER	2001-02-14	M	201	201	COMPUTER SC
	12347	ANI	1999-01-11	F	203	201	COMPUTER SC
	12346	VINI	2001-03-12	F	202	201	COMPUTER SC
	12345	HAMEN	2001-03-24	M	201	201	COMPUTER SC
	12348	PETER	2001-02-14	M	201	202	INFN SC
	12347	ANI	1999-01-11	F	203	202	INFN SC
	12346	VINI	2001-03-12	F	202	202	INFN SC
	12345	HAMEN	2001-03-24	M	201	202	INFN SC
	12348	PETER	2001-02-14	M	201	203	CSE
	12347	ANI	1999-01-11	F	203	203	CSE
	12346	VINI	2001-03-12	F	202	203	CSE
	12345	HAMEN	2001-03-24	M	201	203	CSE
	12348	PETER	2001-02-14	M	201	204	MECHANICAL
	12347	ANI	1999-01-11	F	203	204	MECHANICAL
	12346	VINI	2001-03-12	F	202	204	MECHANICAL
	12345	HAMEN	2001-03-24	M	201	204	MECHANICAL

5. EQUI JOIN

SELECT E.ENO, E.NAME, E.DOB, E.GENDER, E.DCODE, D.DNAME

FROM EMPLOYEE E

JOIN DEPARTMENT D

ON E.DCODE = D.DCODE;

	ENO	NAME	DOB	GENDER	DCODE	DNAME
▶	12345	HAMEN	2001-03-24	M	201	COMPUTER SC
	12346	VINI	2001-03-12	F	202	INFN SC
	12347	ANI	1999-01-11	F	203	CSE
	12348	PETER	2001-02-14	M	201	COMPUTER SC

6.LEFT OUTER JOIN

```
SELECT E.ENO, E.NAME, E.DOB, E.GENDER, E.DCODE, D.DNAME
FROM EMPLOYEE E
LEFT JOIN DEPARTMENT D
ON E.DCODE = D.DCODE;
```

	ENO	NAME	DOB	GENDER	DCODE	DNAME
▶	12345	HAMEN	2001-03-24	M	201	COMPUTER SC
	12346	VINI	2001-03-12	F	202	INFN SC
	12347	ANI	1999-01-11	F	203	CSE
	12348	PETER	2001-02-14	M	201	COMPUTER SC

7.RIGHT OUTER JOIN

```
SELECT E.ENO, E.NAME, E.DOB, E.GENDER, E.DCODE, D.DNAME
FROM EMPLOYEE E
RIGHT JOIN DEPARTMENT D
ON E.DCODE = D.DCODE;
```

	ENO	NAME	DOB	GENDER	DCODE	DNAME
▶	12348	PETER	2001-02-14	M	201	COMPUTER SC
	12345	HAMEN	2001-03-24	M	201	COMPUTER SC
	12346	VINI	2001-03-12	F	202	INFN SC
	12347	ANI	1999-01-11	F	203	CSE
	NULL	NULL	NULL	NULL	NULL	MECHANICAL

8.FULL OUTER JOIN

```
SELECT E.ENO, E.NAME, E.DOB, E.GENDER, E.DCODE, D.DNAME
FROM EMPLOYEE E
LEFT JOIN DEPARTMENT D
ON E.DCODE = D.DCODE
UNION
SELECT E.ENO, E.NAME, E.DOB, E.GENDER, E.DCODE, D.DNAME
FROM EMPLOYEE E
RIGHT JOIN DEPARTMENT D
```

ON E.DCODE = D.DCODE;

	ENO	NAME	DOB	GENDER	DCODE	DNAME
▶	12345	HAMEN	2001-03-24	M	201	COMPUTER SC
	12346	VINI	2001-03-12	F	202	INFN SC
	12347	ANI	1999-01-11	F	203	CSE
	12348	PETER	2001-02-14	M	201	COMPUTER SC
	NULL	NULL	NULL	NULL	NULL	MECHANICAL

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OBSERVATION	
RECORD	
TOTAL	

RESULT:

Thus a program to implement subqueries is written and executed successfully